No prior knowledge needed. No bullshit.

Bitcoin Beginner Guide

From zero to self-custody. Buy Bitcoin, set up a hardware wallet, secure your seed — step by step, directly actionable.

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Preface

Bitcoin is not a complicated technology. Most people do not fail at understanding it, they fail at getting started. Too many YouTube videos, too many contradictory pieces of advice, too many opinions about the right entry point.

This guide solves exactly that. Eight chapters, one clear path. At the end you will have your first sats in genuine self-custody, secured on a hardware wallet, with a properly stored seed phrase.

What are sats? Sats is short for satoshi, the smallest unit of Bitcoin. One Bitcoin consists of 100,000,000 sats, similar to how one euro consists of 100 cents. You do not have to buy a whole Bitcoin. Even 10 or 50 euros is enough to get started.

What this guide is not: investment advice. Whether, when, and how much you buy is up to you. This guide shows you **how** to do it safely, once you have made that decision.

Everything here is based on one principle: simple enough to understand, precise enough to trust.

This guide is free. **Sharing is explicitly welcome:** pass it on to family, friends and anyone who wants to get started with Bitcoin. Ideally unchanged and complete, so everyone knows where it comes from and where to find the current edition.

Let's go,

Alien Investor

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Chapter 1: Why Bitcoin — and why custody is everything

Before you buy even one sat, it is worth understanding what Bitcoin solves. Not because you need that to make a purchase, but because your actions become more consistent once you understand the foundation.

The problem with money

Traditional money (euro, dollar, any fiat currency) can be created in unlimited quantities by central banks. This does not happen abstractly somewhere in the background, it happens concretely: the purchasing power of your saved money declines, and there is nothing you can do about it. This is called inflation, and it is not an accident. It is a feature of the system.

> **THE MOST IMPORTANT NUMBER**
>
> Bitcoin has a hard limit of **21 million coins**. This value is anchored in the protocol and cannot be changed: not by governments, not by companies, not by developers. There is no authority that could produce more.

What makes Bitcoin different

Bitcoin is digital money with a fixed supply. There is no central bank printing more. The issuance of new Bitcoin follows a known, unchangeable schedule that halves every four years. The last Bitcoin will be mined sometime around the year 2140.

On top of that: Bitcoin is the first system in history that enables genuine digital ownership. No trust in a bank required. No custodian. No "account frozen due to suspicion." You are the sole authority.

Why custody is everything

Here begins the important part that is missing or downplayed in many beginner materials:

Bitcoin on an exchange is not your Bitcoin.

An exchange, whether Coinbase, Kraken, Bitpanda or any other, holds your coins in a shared pool. You have a balance in their database, not a private key to a real Bitcoin address. If the exchange goes bankrupt or gets hacked, your coins are gone.

Exchanges are attack targets

A centralized exchange concentrates the wealth of hundreds of thousands of users in one place. That makes it the most attractive target in the world, for criminal hackers and state actors alike.

> **DOCUMENTED HACKS: NOT THEORY**
>
> **Mt. Gox (2014):** Around 850,000 BTC vanished, about 200,000 of which later turned up again. At its 2013 peak, around 70 % of global trading ran through Mt. Gox. Creditors waited over 10 years for partial repayments.
>
> **Bitfinex (2016):** 119,754 BTC taken through an exploit in the wallet infrastructure, despite supposedly high security standards.
>
> **FTX (2022):** 8 billion dollars in customer funds embezzled internally. An additional 477 million dollars stolen in a separate hack after the collapse.
>
> **Bybit (February 2025):** 1.5 billion dollars taken in a matter of hours, the largest cryptocurrency hack of all time. The FBI confirmed the Lazarus Group as the perpetrators.

The Lazarus Group is a state-sponsored hacker group from North Korea. In 2025 alone, North Korean hackers stole a good 2 billion dollars according to Chainalysis, more than 6.7 billion in total, mostly through targeted attacks on crypto exchanges. Their methods: social engineering against developers, manipulated software updates, compromised infrastructure. The money finances the North Korean regime.

The key point: in each of these attacks, users had their funds on the exchange. Anyone holding on a hardware wallet in genuine self-custody does not carry this exchange risk: a hack of the exchange leaves their coins untouched. Self-custody is not entirely free of trust either, though, as the following case shows.

> **THE COLDCARD CASE (JULY 2026): DID EVERYTHING RIGHT AND STILL GOT ROBBED**

> On 30 July 2026 an attacker emptied 1,196 Bitcoin addresses within 41 minutes, around 1,083 BTC in total, worth a good 70 million dollars at the time. By the end of August, analysts counted around 1,800 BTC from more than 8,000 addresses across all waves. The victims had not used an exchange and had done nothing wrong: hardware wallet, seed backed up offline, no click on a bad link. The flaw was in the seed generation itself. Since March 2021, the firmware of affected Coldcard devices had taken the randomness for new seeds from a predictable software generator instead of the built-in hardware random number generator. On the older Mk2 and Mk3 models, this made the number of possible seeds so small that attackers could try them offline and match them against the public blockchain. On the Mk4, Q and Mk5 the weakening was smaller, but still serious. A seed from weak randomness looks like any other, so the owners had no way of noticing. Only seeds into which the user had entered at least 50 private dice rolls of their own were not at risk. A strong passphrase was an extra barrier, but it does not repair the seed.

Self-custody removes the exchange risk, but not your trust in the maker of your device. The randomness behind the seed is the root of every wallet, and you can produce it yourself, with dice. That is exactly why I wrote Volume 4 of the series, *Your Own Bank*: it opens with this case and shows how to generate and verify a seed yourself instead of blindly trusting the device (more at the end of Chapter 5). Anyone who ever generated a seed on a Coldcard with affected firmware without entering at least 50 private dice rolls of their own should treat it as compromised, even if it now sits on another device or has a passphrase: generate a new seed, on a Coldcard only after the firmware update, and move the funds.

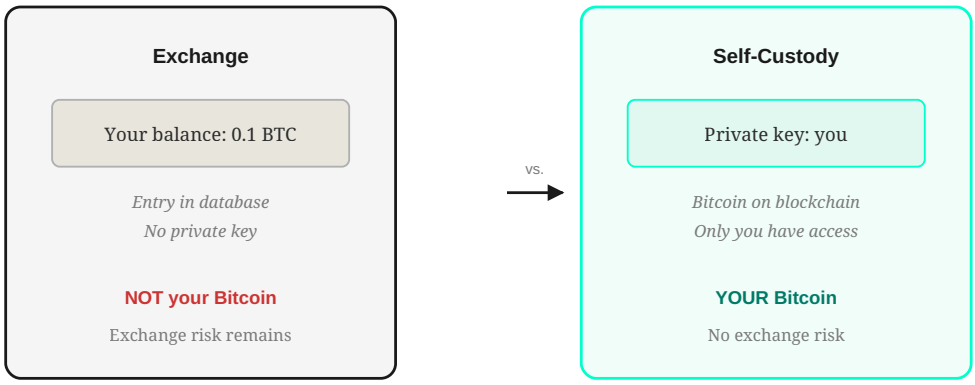


FIG. 01 — Exchange balance vs. true self-custody. Only whoever controls the private key controls the coins.

The phrase that has established itself in Bitcoin circles summarizes it precisely: **"Not your keys, not your coins."** It sounds like a slogan, but it is a technical fact.

This guide leads you to the real solution: a hardware wallet where you control the keys yourself.

Chapter 2: Buying Bitcoin — with and without KYC

If you want to buy Bitcoin, there are two fundamentally different paths: with identity verification (KYC) or without (noKYC). Both have their place. Which you choose depends on your priorities.

What does KYC mean?

KYC stands for "Know Your Customer," a regulatory requirement where you verify yourself with an ID document and possibly a selfie. Regulated exchanges in Europe are required to do this. Your purchase history is documented and stored with a company.

noKYC means: you buy Bitcoin from person to person (peer-to-peer) or via platforms without identity verification. Your purchase history is not stored centrally.

> **NO JUDGMENT**
>
> Whether you buy KYC or noKYC is not a moral question. KYC exchanges are the most practical entry point for many people. noKYC requires more effort but offers more privacy. Both paths lead to real Bitcoin.

> **THE RIGHT MINDSET BEFORE YOUR FIRST BUY**
>
> **Only invest what you can afford to lose.** Bitcoin is volatile, double-digit daily moves are normal. Anyone using money they need for next month's rent makes bad decisions under pressure.
>
> **A savings plan beats timing the market.** Almost nobody catches the perfect entry point. Buying a fixed amount regularly (dollar-cost averaging) smooths the price over time and takes the emotion out of it. Better 25 euros every week than everything at once.
>
> **Do not panic-sell.** The most common beginner mistake is selling in a crash and buying back at the top. Bitcoin is a long game. "Time in the market beats timing the market."

Path 1: Regulated exchange with KYC

For getting started, European exchanges with a good reputation and reasonable fees make sense. Our first recommendation:

Buy Bitcoin in Europe — 21bitcoin

Bitcoin-only app from Europe, simple entry point and DCA, ideal for stacking sats regularly instead of leaving everything in depreciating euros.

With the code **ALIENINVESTOR** you get a permanent 0.2 percentage point fee reduction on instant and savings plan purchases.

alien-investor.org/21bitcoin

Kraken

kraken.com · Withdrawal to hardware wallet supported

One of the oldest and most solid providers (since 2011). Good reputation, MiCA-licensed (Central Bank of Ireland), approved EU-wide including Germany, SEPA transfer supported. Fees depending on volume and trade type.

Bity

bity.com · Swiss provider

Swiss Bitcoin broker. Simple interface, direct purchase, SEPA. Suitable for smaller purchases without much prior knowledge.

Relai

relai.app · Bitcoin only, withdrawal only · MiCA-licensed

App-based, Bitcoin-only. Buys automatically and sends directly to your wallet address. No account on the platform, no custodial risk. MiCA-licensed since October 2025 with Instant SEPA available. Over 500,000 app downloads (as of October 2025).

Important with every exchange: Buy Bitcoin there, but get into the habit from the start of transferring your sats to your hardware wallet promptly. For very small amounts it is fine to leave them on the exchange briefly, but make it a habit to move them right away.

Anyone who waits until they have "enough" may one day face an unpleasant surprise. The exchange is where you buy, not where you store.

Path 2: Buying noKYC

Anyone who prioritizes privacy when buying uses peer-to-peer platforms. That means more effort, a slightly higher spread, but no centralized purchase history. noKYC is also well suited for buying small amounts of sats directly over the Lightning Network, for example to pay on **Nostr** or use other Lightning services.

> **WHAT IS NOSTR?**
>
> Nostr is a decentralized, censorship-resistant social network, similar to Twitter/X but without central control. Users pay each other small sat amounts ("zaps") as tips for good posts. For these micropayments over Lightning, small noKYC amounts are ideal.

Peach Bitcoin

peachbitcoin.com · Mobile app, Europe-focused

P2P platform for Europe. Buy via SEPA or other methods from other users. Simple interface, no account creation required. Good entry point for noKYC. Without KYC up to 1,000 CHF per day (per month for P2P payment methods) and 100,000 CHF per year, 2% buyer fee (flat). Operated under Swiss law.

Bisq 2 / Bisq Easy

bisq.network · Desktop client, decentralized

Fully decentralized, no server, no operator. Runs as a local application on your own machine. With **Bisq 2 / Bisq Easy**, buyers do not need to post a security deposit, ideal for the noKYC entry point with small amounts, especially when you do not own any Bitcoin yet. The older version **Bisq 1** allows larger amounts but requires the buyer to lock up a security deposit in BTC. If you have no coins yet, that leaves you in a chicken-and-egg problem. This is exactly what Bisq Easy is built for: getting your first sats without having to put up any BTC. **Note:** Bisq 1 had a security incident in May 2026 that was fixed in version 1.10.0. If you use Bisq 1, only run the current version.

RoboSats

robosats.com · Lightning-native, federation model

P2P platform for small amounts over Lightning Network. Tor-only, no registration, highly anonymous. Thanks to the federation model (multiple independent coordinators), particularly resilient against shutdown. Interesting for later steps (Chapter 7).

Mostro

mostro.network · Nostr-native, Lightning, experimental

Decentralized P2P exchange running directly on the Nostr protocol. No KYC, no account, identity is based on Nostr keys. Bitcoin is secured via Lightning hold invoice until fiat payment is confirmed. Mostro Mobile for Android, plus desktop versions, both via github.com/MostroP2P/mobile, and a command-line client. **Still experimental:** suitable only for small amounts and users with basic technical knowledge.

Step by step: First purchase on an exchange

1 **Create and verify account**

Register with your chosen exchange. The KYC process takes 10–30 minutes depending on the provider. Have your ID document, possibly a selfie, and proof of address ready.

2 **Deposit funds**

SEPA transfer is the cheapest option in Europe. Copy the exchange's IBAN from your account, do not forget the reference number.

3 **Buy Bitcoin**

Use a limit order instead of a market order: you enter a desired price and wait. A market order buys immediately at the current rate, often with a worse spread.

4 **Leave on the exchange for now**

For small amounts it is fine to leave the sats on the exchange briefly. Still, set up your hardware wallet promptly (Chapter 3) and transfer your coins to your wallet (Chapter 4) before buying more. The sooner you move them, the better.

Step by step: First noKYC purchase with Bisq Easy

Bisq Easy is the simplest entry protocol of Bisq 2: no security deposit, no prior knowledge required, no fees for buyers. Suitable for first purchases up to approximately 600 USD.

1 Download Bisq 2

Visit bisq.network and download the desktop app for your operating system. Verify the signature — instructions on the download page. Bisq runs locally on your computer, there is no central server.

2 Start the Bisq Easy protocol

In Bisq 2, open the **Buy Bitcoin** section and select the **Bisq Easy** protocol. Enter currency EUR and choose a payment method: **SEPA** or SEPA Instant for Germany/Austria, Twint for Switzerland.

3 Select an offer

A list of active sellers appears, sorted by price and reputation score. Prefer sellers with a high score and many completed trades. Enter the amount and provide your Bitcoin receiving address (hardware wallet).

4 Send SEPA transfer

Bisq Easy shows the seller's IBAN. Send the transfer with exactly the reference stated. **Important:** copy the reference character by character. Incorrect details can jeopardize the trade.

5 Confirm payment

After sending the transfer, click **"Payment sent"** in Bisq Easy. The seller is notified and confirms once payment arrives. SEPA standard: 1–2 business days. SEPA Instant: minutes.

6 Receive Bitcoin

Once the seller confirms receipt, the sats are transferred directly to your hardware wallet address. Trade complete: no KYC, no account, no data on file.

So Bitcoin can be bought without any exchange at all, directly from person to person. This is not a fringe route but the very core of what makes Bitcoin free: no middleman who can

verify, freeze, or watch you. If you want to go deeper, there are two detailed guides on the blog.

DEEP DIVE · BUYING NOKYC

Bisq step by step & paying anonymously

The **buying guide** walks step by step through a real Bisq Classic trade (SEPA, security deposit, multisig) and uses Bisq Easy to solve the chicken-and-egg problem of the deposit. The **payment methods** article compares which route is truly private, from SEPA through Cash by Mail to the Monero route.

BUYING GUIDE
alien-investor.org/en/bisq-kaufen-anleitung.html

PAYMENT METHODS
alien-investor.org/en/bisq-zahlungsmethoden-anonym.html

Chapter 3: Setting up a hardware wallet (BitBox02)

A hardware wallet is a small device that generates and stores your private key offline. The key never leaves the device, so even if your computer is compromised, your coins are safe. For anything above a few hundred euros, a hardware wallet is mandatory.

This guide walks through the process using the **BitBox02 (Bitcoin-only Edition)** by Shift Crypto from Switzerland. It is easy to set up, open-source, and widely used in Europe. The same principles apply to other hardware wallets (Coldcard, Jade, Foundation Passport).

Hold your own Bitcoin — BitBox hardware wallet

"Not your keys, not your coins." With the BitBox you hold your sats yourself. Made in Switzerland, open-source, easy to set up.

5 % discount with code **ALIENINVESTOR** in the official shop.

alien-investor.org/bitbox

BitBox02 Nova — for Apple users (iPhone/iPad)

The BitBox02 connects via USB-C, which does not work directly with iPhones or iPads. The **BitBox02 Nova** is the solution: it connects via **Bluetooth** to the BitBoxApp and is fully compatible with iOS. Same security architecture, open-source, made in Switzerland. Available in Midnight Black and Polar White, the Bitcoin-only edition also in Bitcoin Orange.



5 % discount with code **ALIENINVESTOR** in the official shop.

shop.bitbox.swiss · ref=alien-investor

BUY HARDWARE WALLETS ONLY FROM THE MANUFACTURER

Never buy a hardware wallet on Amazon or from third parties, not even supposedly factory-sealed. The risk of tampering is real. Always order directly from the manufacturer: **bitbox.swiss**



This is how your BitBox02 should arrive: originally sealed in vacuum packaging with the BitBox logo intact. If the seal is opened or damaged, do not use the device and contact the manufacturer.

What you need

BitBox02 (Bitcoin-only Edition), USB-C cable or adapter (included), Mac, Windows or Linux computer, and the **BitBoxApp** from bitbox.swiss (free).

Setup step by step

1 Download the BitBoxApp

Go to **bitbox.swiss/start** and download the app for your operating system. Verify the signature of the download (instructions on the download page). This protects against tampered installation files.

Linux users: the .deb and .rpm packages set up the udev rules themselves. If you use the AppImage, copy `53-hid-bitbox02.rules` and `54-hid-bitbox02.rules` to `/etc/udev/rules.d/` by hand, otherwise the device will not be recognized. Instructions on the BitBox support site.

> **PRO: BITBOXAPP AS APK VIA OBTAINIUM (ANDROID)**
>
> Anyone running GrapheneOS or who prefers to get apps directly from the developer rather than from the Play Store can obtain and keep the BitBoxApp up to date via **Obtainium**. Obtainium downloads APKs directly from GitHub and checks for updates automatically. **AppVerifier** then compares the cryptographic fingerprint of the app against the known certificate, so even the first download is verified.
>
Source	`github.com/BitBoxSwiss/bitbox-wallet-app`
Regex	`.*-android\.apk$`
Package	`ch.shiftcrypto.bitboxapp`
Certificate fingerprint	`67:E9:05:08:F7:49:BE:9D:CF:B1:83:33:92:BA:99:EE:BF:89:37:B1:BA:DD:F3:05:63:1C:E2:D9:FB:A5:0B:D7`
>
> *These details are accurate at time of printing. File names, regex, and fingerprints can change if the developer updates their build pipeline. Always verify yourself: check the current release on GitHub and compare the fingerprint against the official developer page.*

2 Connect device and check pairing

Connect the BitBox02 via USB-C, open the BitBoxApp. The device is recognized, firmware updates are offered automatically, always keep it current.

Check the pairing code: The app shows a code that must be identical to the code on the BitBox display. If they do not match, disconnect the device immediately: this is a tamper warning signal.

Android: If the device is not recognized, enable OTG in settings. **Apple iOS:** Only possible with the BitBox02 Nova via Bluetooth.

3 Create a new wallet with recovery words

Do not tap "Create wallet" yet, tap "Advanced options" below it. By default the wizard would write the backup to a microSD card. For this guide, turn on "Skip microSD card backup and write down recovery words instead" there and leave "Create 12-word instead of 24-word seed" off. Only then tap "Create wallet". **Do not restore an existing wallet.** You are starting fresh.

4 Set the device name and device password

The wizard first asks for a name for the device, then for the device password. Set it directly on the device. It protects the BitBox against physical access and is not the same as the passphrase from Chapter 6. Then the device generates the seed phrase itself, only the device knows the key.

5 Write down the seed phrase (Chapter 5 applies here)

The device shows you 24 words. Write them down by hand on paper. **Not digitally, not yet.** The order matters. Chapter 5 explains how to secure them permanently.

Absolute security rules at this moment: No cameras in view, no witnesses. Never say the words out loud, microphones can be anywhere. Put your smartphone in another room.

6 Confirm the seed phrase

The device asks you to confirm each of the 24 words. This step ensures you have written them down correctly. Only then is the wallet active.

+ Optional: additional microSD backup

The BitBox02 can also create a backup on a microSD card, in the BitBoxApp under "Manage device" at "Manage backups". **This backup is not encrypted:** whoever has the card has the wallet. So store it as carefully as the seed phrase. Only ever insert the card into the BitBox, never into a PC or smartphone, where malware could read the seed. After the backup, take it out and store it separately from the device. The microSD does not replace the seed phrase backup on metal, it is an additional backup with its own risk.

7 Display and test receiving address

Go to "Receive" in the BitBoxApp. The device shows the address, confirm it on the display. Send a small test amount from your exchange to this address. Wait until it arrives. Only then transfer the rest.

After setup, the device is your vault. The coins you send there are under your sole control, as long as your seed phrase is safe.

Chapter 4: Transferring Bitcoin from the exchange to your wallet

The hardware wallet is set up. Now comes the step where most mistakes happen: the first transfer from the exchange to your own wallet. Not because it is technically complicated, but because a wrong address or wrong network can lead to permanent loss. Once you understand it correctly, it becomes routine.

Address types: what you need to know

Bitcoin addresses look different depending on which standard they use. When an exchange asks for an address format:

Native SegWit — starts with `bc1q` ✓ Recommended

BitBox02 standard · lowest network fees · supported by almost all exchanges

This is the address your BitBox02 generates by default. Always choose this when the exchange offers a selection. bc1q addresses store only the **hash** of your public key, not the key itself. That means: as long as you have never spent from this address, your public key is not visible on the blockchain. A quantum computer would need the public key to derive the private key from it. Unused bc1q addresses are therefore largely protected against future quantum attacks, as long as nobody knows your xPub. Once you spend from the address for the first time, the public key becomes visible. After that, do not reuse the address.

Taproot — starts with `bc1p`

Newest standard · not yet supported by all exchanges

Technically more advanced, but not universally available yet. Important difference from bc1q: with Taproot addresses, the public key is embedded directly in the output script, even before you have ever spent from it. A bc1p address with a balance is therefore theoretically more vulnerable to future quantum attacks. For getting started, prefer Native SegWit (bc1q).

Legacy — starts with `1`

Outdated · higher fees

Works technically, but unnecessarily expensive. Like bc1q, Legacy stores only the hash of the public key. Unused legacy addresses are likewise largely protected as long as they have never been spent from and nobody knows your xPub. Only choose this if an exchange exclusively supports this format.

> **CHECK THE NETWORK — CRITICAL POINT**
>
> Many exchanges offer a network selection when withdrawing: Bitcoin, Ethereum, BNB Smart Chain, and others. Always and exclusively choose **Bitcoin (BTC)**. Anyone who sends their sats over a different network loses them permanently. A Bitcoin hardware wallet cannot receive or display coins on another network.

Understanding network fees

Every Bitcoin transaction costs a small fee. It does not go to a company but to the miners who include your transaction in a block. Important to understand: **the fee does not depend on the amount sent**, but on the data size of the transaction, measured in **sat/vByte** (satoshi per virtual byte). Whether you send 50 euros or 50,000 euros makes almost no difference to the fee.

What determines the fee is **network demand**: only a limited number of transactions fit into a block (one roughly every 10 minutes). When many people are active, transactions compete for space, and whoever offers more sat/vByte gets in faster. When it is quiet, you pay a few cents.

> **HOW TO CHOOSE THE RIGHT FEE**
>
> Most wallets and exchanges automatically suggest a sensible fee (often "economy", "normal", "priority"). For a withdrawal to your own wallet you are rarely in a hurry: a low to normal fee is almost always enough. On **mempool.space** you can see the currently recommended sat/vByte values and judge whether it is a good moment for cheap transactions. Tip: weekends are often quieter and therefore cheaper.

First withdrawal: step by step

1 Open the receiving address in the BitBoxApp

Connect the BitBox02, open the BitBoxApp, click **Receive**. The device shows the address, confirm it on the display. Copy the address to the clipboard — never type it manually.

2 Set a test amount

For the first transfer, always start with a small amount, for example 10–20 euros worth. Anyone who starts with their entire balance has no second chance if something goes wrong.

3 On the exchange: open withdrawal

On 21bitcoin: open the Bitcoin section in the app, then select Withdraw. Paste the address by scanning the QR code or from the clipboard. Network: **Bitcoin (BTC)**.

4 Check the address before confirming

Compare the first and last four characters of the pasted address with the address on the BitBox display. If they match, everything is correct. Only then submit.

5 Track the transaction on the blockchain

Bitcoin transactions take a few minutes to an hour depending on network load. You do not have to wait blindly: open **mempool.space** in your browser and enter your receiving address (bc1q...) in the search bar. You can see the transaction immediately after it is sent, even before it is confirmed, along with the current confirmation count. After one confirmation it is secure. Wait until the amount also appears in the BitBoxApp, then withdraw the rest from the exchange.

Auto-Wallet-Transfer on 21bitcoin

Anyone buying regularly does not want to withdraw manually with every purchase. 21bitcoin offers the **Auto-Wallet-Transfer** for this: you set a wallet address and a satoshi threshold once. When your balance reaches that amount, the full balance is automatically sent to your hardware wallet.

> SETTING UP AUTO-WALLET-TRANSFER

> **Where in the app:** Bitcoin symbol at the bottom of the tab bar, then select *Auto Transfer*.
>
> **Threshold:** Configurable by you. 21bitcoin recommends at least 1,000,000 sats to avoid tiny amounts (dust). You can also set a lower value, for example from 200,000 sats (around 150 euros, depending on the rate).
>
> **Fee:** 1,000 sats fixed (less than 1 euro), network fee already included. Maximum amount per transfer is 2.1 BTC.
>
> **xPub or single address?** Instead of a fixed address, you can store the Extended Public Key (xPub) of your BitBox. 21bitcoin then generates a new address from your wallet for each transfer, which is better for your privacy on-chain. However: anyone who knows your xPub can derive all future addresses of your wallet from it and observe all incoming transactions. 21bitcoin thereby gets more visibility into your wallet than with a single address. You find the xPub in the BitBoxApp in the account under *Account information → Extended public key*.
>
> **Alternative: single address manually.** You can also simply copy a bc1q address from the BitBoxApp and store it at 21bitcoin. All deposits from 21bitcoin go to this address. When you want to change it later, generate a new address in the BitBoxApp and enter it. 21bitcoin sees only one address instead of your entire wallet tree. The downside: address reuse is visible on-chain, multiple deposits to the same address can be linked by observers. It is a trade-off: less data at 21bitcoin, but more on-chain transparency.

With the Auto-Wallet-Transfer, 21bitcoin operates in non-custodial mode: you buy, and the sats land automatically on your hardware wallet. The exchange never holds your balance longer than necessary.

Chapter 5: The seed phrase — the most important thing

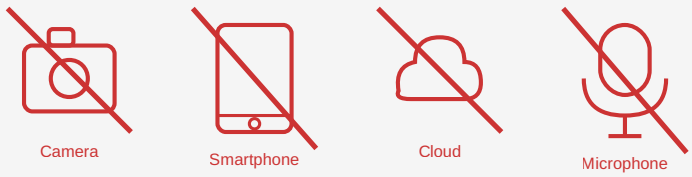
Your hardware wallet is replaceable. Your seed phrase is not. If the device breaks, gets stolen, or is lost, you can restore any compatible wallet with the seed phrase and immediately regain access. Without the seed phrase, your coins are lost forever.

No excessive alarm: millions of people use hardware wallets without problems. But handling the seed phrase is the one area where no mistake is acceptable.

What is a seed phrase?

A seed phrase is a sequence of 24 (or sometimes 12) English words from a defined list of 2,048 words (BIP-39 standard). These words encode your master key. All private keys and addresses of your wallet are deterministically derived from them.

> **DIGITAL SECURITY TRAP**
>
> Never: screenshot, photo, cloud storage (iCloud, Google Drive, Dropbox), password manager, email, messenger, digital document. Anyone who sees the words digitally can steal your coins. This applies to supposedly secure services too.



Paper: the first step

Directly during wallet creation, you write down the 24 words by hand on paper. Numbered, legible, correct order. This is your first draft, not the final solution.

Paper burns, rots, fades. For long-term storage you need metal. Anyone looking for a clean intermediate step can use the **BitBox Backup Card**: a laminated card from Shift Crypto, specifically designed for seed phrases, water-resistant, sturdy, with a clearly structured layout for all 24 words.

BitBox Backup Card

Laminated card for secure seed phrase storage. Water-resistant, structured for all 24 words. A clean first step before metal backup.

5 % discount with code **ALIENINVESTOR** in the shop.

shop.bitbox.swiss · ref=alien-investor



Tamper-Evident Bags

Tamper-evident sealing bags for your seed phrase backups. Once sealed, it is immediately visible if someone has opened the bag, ideal for storing backup cards or paper copies securely and ruling out unnoticed access.

5 % discount with code **ALIENINVESTOR** in the shop.

shop.bitbox.swiss · ref=alien-investor



Metal: the proper backup

Metal plates are the best method for long-term storage. They survive typical house fires (approx. 800–1000°C), water, and mechanical force. There are several variants:

Steelwallet Pro / Seedor

shop.bitbox.swiss · ref=alien-investor · High quality, letter-based

Letter tiles or entire stamped word discs are inserted into a steel housing and secured. Elegant and compact. Two variants: Steelwallet Pro (tiles in grooves with spring locks, no tools) and Seedor (stamping set with hammer).

Steelwallet (BitBox)

shop.bitbox.swiss · ref=alien-investor

Steel plate for punching in the seed words. Cheaper than Cryptosteel, functionally identical. Includes a punch tool. Directly from the BitBox manufacturer.

DIY

Stainless steel sheet, punch, hammer

Punch and stainless steel sheet from a hardware store. Costs less than 20 euros, lasts centuries. Instructions available from Glacier Protocol (free).

Steelwallet Pro — stainless steel tile system

Flat stainless steel cassette with grooves: insert the letter tiles, a spring lock at the end of each groove holds them in place, done. No hammer, no tools required, and reusable if your setup ever changes. Compact, made of high-quality stainless steel, built to resist fire, water and physical impacts.



Why only four letters per word? In the BIP-39 standard, all 2,048 words are chosen such that the first four letters of each word are unique. The Steelwallet Pro uses exactly that: you only set the first four letters, which is sufficient for full recovery. Wallets such as the BitBox even auto-complete the word as soon as you enter the first characters.

5 % discount with code **ALIENINVESTOR** in the shop.

shop.bitbox.swiss · ref=alien-investor

Seedor — stamping set with hammer

Stamp letters into stainless steel plates with a hammer, simple, permanent, affordable. Includes hammer, letter stamps, and positioning frame. Good for those who prefer a hands-on approach.

Unlike the Steelwallet Pro, here the **full words** are stamped in. The individual metal plates are then threaded onto a pin and screwed together, which makes the system exceptionally robust. A further advantage: should the cylinder deform or come apart under extreme conditions, all words are still present individually and remain readable. Each plate is also **numbered**, so the correct order can always be reconstructed unambiguously.

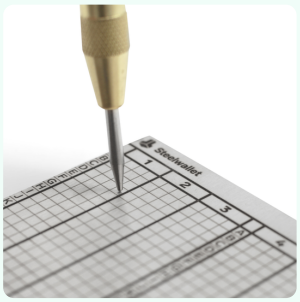


5 % discount with code **ALIENINVESTOR** in the shop.

shop.bitbox.swiss · ref=alien-investor

Steelwallet — metal backup from the BitBox manufacturer

Stainless steel plate for punching in all 24 seed words. Fireproof, waterproof, lasts decades. Includes punch tool, ready to use without searching for tools at the hardware store.



5 % discount with code **ALIENINVESTOR** in the shop.

shop.bitbox.swiss · ref=alien-investor

The 3-2-1 principle

Adapted from data backup practices, it works equally well for seed phrases:

01 **3 copies** of your seed phrase exist

02 on **2 different media** (e.g. metal and paper)

03 with **1 copy at a different location.** But be careful: banks and safe deposit boxes can be frozen, seized, or simply inaccessible when you need them. If you store anything there, keep only a **part** of your secret, for example only the optional passphrase (more on this in Chapter 6) stored without any recognizable context, while the seed itself is kept elsewhere. Or the reverse: the seed without passphrase in the safe deposit box, the passphrase at home. Decide consciously what is kept where. Your complete secret must never be concentrated in a single location.

> **BANKS ARE NOT A SECURE SOLUTION**
>
> Bank accounts can be frozen, safe deposit boxes seized, and institutions can become insolvent. This is not theory, it is history. Anyone storing their complete seed in a bank safe deposit box has delegated custody back to a third party. Use banks at most as one of several locations for a **part** of the secret, and keep it unidentifiable: no label, no obvious context.

Where NOT to store

At home in a drawer: too easy to find. In a safe: good, but a single copy is not enough. At a bank: the bank could be frozen, leaving you without access. Together with the hardware wallet: if stolen, you lose both at once.

Separating the device from the seed phrase is critical. Important to understand: an attacker does not need the hardware wallet at all. Anyone who knows the complete seed phrase (and the passphrase, if one is set) has full control over all sats, without ever touching the device. Any compatible wallet software can restore the key from the seed phrase. The device protects the seed phrase during normal use, but does not replace the secure storage of the seed itself. Whoever has the seed has everything.

TEST AFTER SECURING — PRACTICE RECOVERY

The most reliable test: reset the device and immediately restore from the seed phrase. This sounds drastic, but it is completely safe: right after setup there are no coins on the wallet yet. In the BitBoxApp, go to "Manage device", choose "Factory reset" under "Expert settings", then select "Restore from recovery words" and enter your 24 words. If the same receiving address and the same root fingerprint appear as before (more on this in Chapter 6), the backup is correct. This lets you practice the real scenario without risk and confirms that your seed phrase works.

DEEP DIVE · BOOK · VOLUME 4 OF THE SERIES

Your Own Bank: Master Bitcoin Self-Custody

Once your first seed phrase is safe, the next questions follow: How do I generate a seed myself without blindly trusting the device? When is a passphrase worth it? How do I spread the risk with multisig? And how do my heirs get to the coins without anyone else being able to? This book from my series walks through these steps in detail. It is my own book, so the recommendation is not neutral.

ALL BOOKS IN THE SERIES

alien-investor.org/en/buecher.html

Chapter 6: The passphrase (25th word) — when does it make sense?

In addition to the 24-word seed phrase, there is an optional extension: the passphrase, sometimes also called the "25th word." It is not a word from the BIP-39 list, but any string of characters you choose yourself.

Technically, the passphrase is an additional input in the key derivation process. Without a passphrase you get Wallet A. With a passphrase you get Wallet B, with completely different addresses, completely different coins, derived from the same 24 words.

What it provides

Three use cases:

Protection against physical compromise: If someone finds your seed phrase, they still do not have access to your coins, they also need the passphrase. The seed phrase alone only opens the wallet without the passphrase. This is the "duress wallet" approach: a small amount on the seed-only wallet, real coins behind the passphrase.

Plausible deniability: You can credibly claim that the seed-only wallet is your only one, while your real holdings are behind the passphrase. To be honest: against an attacker who knows that passphrases exist and can apply pressure, this decoy is weak protection. It helps against someone who finds your backup, not against an informed extortionist.

Multiple independent wallets from one seed phrase: Each different passphrase creates a completely separate wallet, with different addresses, different coins, fully independent from each other. This makes it possible, for example, to manage multiple pools with a single seed phrase: one wallet for regular spending, one for long-term savings, one for shared family funds. Or each family member gets their own passphrase, each having their own wallet, all based on the same seed phrase. Anyone who knows the seed but not the individual passphrase still cannot access that wallet.

What it costs

> **THE PASSPHRASE CANNOT BE RECOVERED**

> If you forget the passphrase, your coins on the passphrase wallet are lost forever. There is no recovery, no backup option, no support. The passphrase must be secured just as carefully as the seed phrase, and separately from it.

That means: now there are two things you must secure and keep in separate locations. The added complexity is real. For beginners who have just understood seed phrase security, the passphrase is not strictly necessary.

When the passphrase makes sense

You hold a significant amount of Bitcoin and want protection even in the event your seed phrase is compromised. You understand exactly what the passphrase is, and you have a clear plan for how and where to secure it.

For the first steps: set up the basic structure correctly first, transfer your first amount, confirm everything works. The passphrase is an optional next step, not a required component.

If you use it: how to do it right

1 **Choose a passphrase**

The safest option, and the easiest to back up, is six dice-rolled words from the EFF long word list (7,776 words, five rolls per word), all in capital letters, separated by single spaces. That gives around 77 bits, can be checked character by character on the device and stamped into steel. Nothing made up, no name, no date, no password you use elsewhere. No accented letters, preferably no special characters. If a word contains a hyphen (such as T-SHIRT), re-roll just that word. Stay below 50 characters so the passphrase also works on a Trezor, and if it comes out longer, re-roll the whole passphrase.

2 **Store separately from the seed phrase**

Never keep the passphrase together with the seed phrase. An attacker who finds both has everything. Separate metal medium, separate storage location.

3 Verify the passphrase wallet

Enable the passphrase feature in the BitBoxApp ("Settings" → "Manage device" → "Expert settings" → "Passphrase") and enter the passphrase on the BitBox02 when unlocking it. Then note the first receiving address, it proves you have the passphrase correct. For full verification: send a small test amount (e.g. a few sats) to this address. Then disconnect the BitBox, close the app, reconnect the device, and enter the passphrase again. If the amount is still there and the receiving address is identical, you are confirmed to be in the correct wallet. This way you know not only that the passphrase is correct, but also that you can enter it reproducibly.

Note the fingerprint: Every combination of seed and passphrase has its own fingerprint, eight characters from the digits 0 to 9 and the letters a to f. The BitBoxApp shows it as "Root fingerprint" in the "Account information" of every account and under "Settings" at "Manage device" in the "Device information" section. Write it down with your passphrase backup. A typo in the passphrase produces no error message, it silently opens a different, empty wallet. If the fingerprint matches your note when you enter the passphrase later, you are in the right wallet. Neither a key nor an address can be derived from the fingerprint. It still does not belong in a chat window or a cloud folder.

4 Inform heirs or create an emergency plan

If you die or become incapacitated, someone must know that a passphrase exists and how to find it. Without this plan, your coins are lost. Write a sealed set of instructions for trusted persons.

The instructions should state that a passphrase exists, where it is kept, and the fingerprint from step 3. Anyone who leaves out the passphrase lands in the wallet without it, and with the decoy approach described above there is a small amount there. A visible balance therefore does not prove it is the right wallet. Only the fingerprint shows that for certain. Volume 4, *Your Own Bank*, explains how to pass on Bitcoin step by step, up to an inheritance path with a timelock.

Estate Planner: printable forms for your heirs, free and offline

I built this Estate Planner for Volume 4. It produces printable forms for your heirs: a proof of existence that may lie in the open, and confidential access instructions with a wallet inventory, locations and the order in which to proceed. It is a single HTML file that runs offline in the browser, with no server and nothing stored automatically, and it is free. It never asks for seed words, a passphrase or access passwords. Anything that would reveal a hiding place is printed as a blank line for you to fill in by hand. Its legal and tax notes cover German inheritance law and are deliberately kept in German.

ESTATE PLANNER & INSTRUCTIONS

alien-investor.org/en/nachlassplaner.html

Chapter 7: What comes next

You have Bitcoin in self-custody. The hardest part is done. What follows now are optional extensions that make sense or not depending on your goals.

Full node: why and when?

A Bitcoin full node is a computer that downloads the entire Bitcoin blockchain, verifies every single transaction since the first block (January 3, 2009) against all consensus rules, without trusting anyone. Not an exchange, not a block explorer, not an API provider. This is the technical implementation of "Don't trust, verify."

Anyone using Bitcoin without their own node is asking third-party servers whether a transaction arrived, what the current fee should be, and how much Bitcoin is at an address. These servers could deliver manipulated information or simply shut down. With your own node you verify yourself, directly against the blockchain.

> **WHAT A FULL NODE CONCRETELY PROVIDES**
>
> **Privacy:** The node downloads all transactions without exception. An observer cannot mathematically determine which ones concern you. Wallet queries run locally, no network packet leaves your machine.
>
> **Censorship resistance:** Transactions are broadcast directly into the P2P network, no third party can stop or filter them.
>
> **Verification:** With a single command you can verify the total supply of all Bitcoin ever created yourself. No block explorer, no exchange can change the 21-million rule without your node immediately rejecting those blocks.
>
> **Political participation:** Full nodes enforce the consensus rules, not miners, not exchanges. The Blocksize War of 2017 proved it: when an overwhelming majority of nodes rejected a planned protocol change, even miners with 80 % of the hash rate backed down. Nodes are the final authority.

To start with, a full node is not essential: first set up self-custody, secure your seed phrase, get comfortable with the system. But the more Bitcoin you hold and the more seriously

you take the subject, the more important your own node becomes. The barrier to entry is low today.

Hardware: what you need

As of mid-2026 the Bitcoin blockchain is around 750 GB and grows by approximately 6–8 GB per month. Plan for at least 2 TB of storage so the drive does not fill up within a few years, and a dedicated address index needs extra space on top. An SSD is mandatory. Without one the node is not practical. NVMe is preferable to SATA.

On memory: 8 GB of RAM is the minimum for a plain Bitcoin node. If you also run a Lightning node, a mempool explorer, an Electrum server and further apps in parallel, go for **16 GB**. With six or seven services running at once, 8 GB quickly hits its limit.

Raspberry Pi 5 (16 GB) + NVMe SSD ⭐ Recommended

5–15 watts · IBD (initial sync): 1–3 days

Best price-to-performance ratio. Quiet, energy-efficient, compact. The 16 GB variant has enough headroom when Lightning and further apps run alongside Bitcoin Core. Ready to run in a few hours with Umbrel or RaspiBlitz.

Mini PC (Intel N100 or similar)

More performance at similar power consumption to the Pi 5

Slightly more computing power at similar consumption to the Pi 5. A good choice if you also want to use the node for other services (Lightning, storage, VPN).

Used business PC (Dell Optiplex, Lenovo ThinkCentre Mini)

Cheap to buy · but: 50–100 watts consumption

Cheap to buy, but older desktop PCs consume significantly more power, which drives up running costs over the years. Only worthwhile if the machine is running anyway or already available.

Software: which distribution?

Umbrel ⭐ Recommended for beginners

umbrel.com · umbrelOS · Easiest entry

Elegant web interface, over 300 apps (Bitcoin Core, Lightning, storage, VPN and more), active community. Setup in minutes. Runs on Raspberry Pi 4/5 or custom hardware. Umbrel Home available as a ready-made plug-and-play device. The best choice for most beginners.

Start9 (StartOS)

start9.com · Privacy-focused, Tor as an add-on

All services run in isolated containers. Since StartOS 0.4 (July 2026), Tor is an optional add-on. More strongly focused on sovereignty and security than Umbrel. Ready-made hardware (Server One/Pure) available, DIY setup requires somewhat more technical knowledge.

RaspiBlitz

raspiblitz.org · Open source, terminal-based

Fully open source, supports NVMe boot on the Raspberry Pi 5 and offers an LCD display for status information. For users with Linux knowledge who want to understand what is happening in the background.

> **FULL NODE GUIDE**
>
> I wrote a detailed article covering all ten reasons to run your own full node, including concrete attack vectors without a node, the history of the Blocksize War, and a detailed setup guide. Find it at:
> **alien-investor.org/en/bitcoin-full-node.html**

Taxes: what you need to know

In Germany, Bitcoin is classified as a private disposal transaction (§ 23 EStG). The most important rule: anyone who holds Bitcoin for more than one year and then sells pays no tax on the gain. Anyone who sells earlier must pay tax on the gain. Transferring between your own wallets, for example from the exchange to your hardware wallet, is not a taxable event.

This applies to Germany. Austria and Switzerland have their own rules. For other countries, consult local tax guidance.

Two things are frequently misunderstood:

FiFo method: The principle is individual lot tracking of each coin. Where that is not possible, the tax authority accepts FiFo ("First In, First Out") as a common simplification per wallet: what you bought first is treated as sold first. Anyone who buys from multiple brokers and transfers between wallets quickly loses track of which coins were bought when and whether the one-year holding period applies.

The exemption threshold is not a free pass: From 2024, a threshold of 1,000 euros per year applies to taxable gains from private disposal transactions. But: this is not a tax-free allowance. Anyone who reaches or exceeds the threshold pays tax on the entire gain, not just the portion above it.

BTC Steuertool — free, open source, computes locally

I built this tool for my own tax return. It reads the CSV exports from 21bitcoin, Bison, Swissquote, Strike, Pocket and Bisq as well as the wallet exports of the BitBoxApp, calculates FiFo including the holding period, and produces a finished report with documentation for the tax office or your tax advisor. It also reads exports from Sparrow, Electrum, Trezor Suite and Ledger Wallet, and other exchanges via a bulk export from CoinTracking or Blockpit. These formats have not yet been confirmed against real exports, though, and the report says so. Coldcard, Jade or SeedSigner work via the Sparrow export, Coldcard and Jade also via Electrum (as of October 2026). noKYC purchases (Bisq, RoboSats, P2P) can be entered and stay in a separate, internal report. No cloud, no registration, no third-party servers.

Completely free since October 2026. The recommended way is the app for Android and GrapheneOS or the Flatpak for the Linux desktop: neither has network permission, and the calculation engine and price data are built in. If you would rather not install anything, use the web version. It comes with sample data to try out and also computes locally in the browser, but fetches its program code from the server on every visit.

Your broker or wallet is missing? Help add it. Export the CSV file, delete every row except the header and a single transaction, and replace amounts, date, addresses and IDs in it with made-up values in the same format. Send that to kontakt@alien-investor.org. The format has to be real, the values do not. Such an export is the release criterion for every new parser, and that also applies to the formats above that are not yet confirmed. Every new parser is then available to all users for free.

APP, FLATPAK & WEB VERSION

alien-investor.org/en/btc-steuertool.html

DEEP DIVE · BOOK · VOLUME 3 OF THE SERIES

The Decentralized System: Bitcoin. Stocks. True Ownership.

How taxes, reporting duties, your own node and Lightning fit together, and how the same principle of true ownership works for stocks and precious metals, is the subject of this book from my series. It is my own book, so the recommendation is not neutral.

ALL BOOKS IN THE SERIES

alien-investor.org/en/buecher.html

This is not tax advice. Rules vary by country and may change. When in doubt, consult a tax advisor familiar with cryptocurrency.

Lightning Network: Bitcoin for everyday use

The Lightning Network is a second layer on top of Bitcoin. It enables fast and cheap transactions without recording every payment on the main blockchain. Good for smaller amounts, micropayments, and everyday use.

Lightning requires a bit more understanding than on-chain Bitcoin. You need to manage liquidity, open and close channels. For getting started, there are simpler ways:

Phoenix Wallet ⭐ Entry point

phoenix.acinq.co · Non-custodial, simple, smartphone

The simplest non-custodial Lightning wallet for smartphones. Channel management runs automatically in the background, you control your keys. Perfect for the first Lightning steps without your own node, just install the app and get started.

Alby Go

albyhub.com · For Umbrel/own node · Maximum control

Anyone running their own Umbrel node with Alby Hub gets the matching mobile app with Alby Go. Connects directly to your own node, all sats remain under your own

control, and setup is still straightforward. The best combination of maximum sovereignty and simple operation for advanced users.

Zeus Wallet

zeusln.app · For node operators, maximum control

Connects to your own Lightning node. Full control over all parameters, but requires a running node and some basic technical understanding.

DEEP DIVE · PAYMENT SOVEREIGNTY

Using Lightning right, without a hidden bank

Custodial wallets like Cash App are convenient, but essentially banks without deposit insurance: the provider holds the keys and can freeze your account. The article explains the difference from non-custodial wallets and the path to your own Lightning node via Umbrel.

LIGHTNING GUIDE

alien-investor.org/en/lightning-network-zahlungssouveraenitaet.html

Further learning: what comes next?

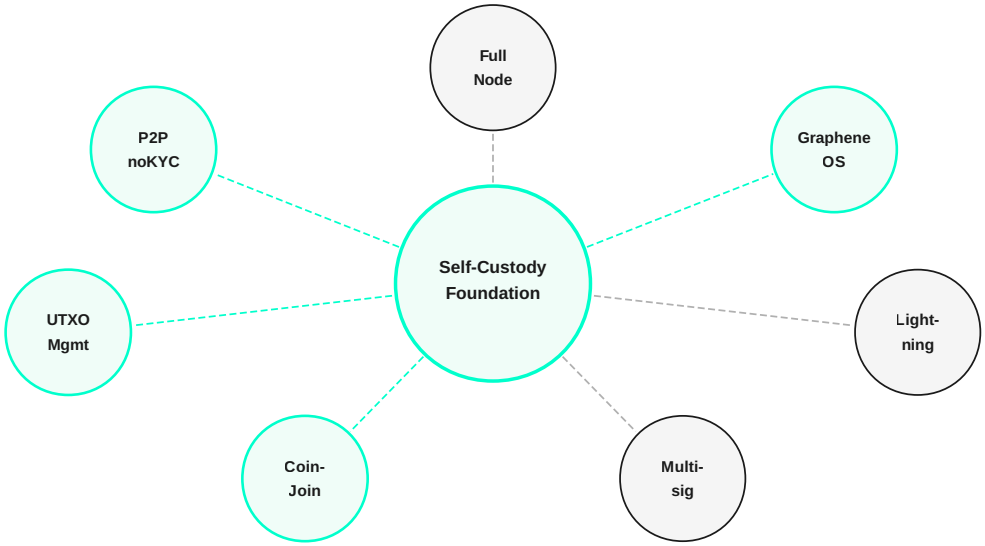


FIG. 02 — Building from the self-custody foundation: Full Node, Lightning and Multisig as infrastructure. P2P buying without KYC, UTXO management, CoinJoin and GrapheneOS for more privacy.

Your smartphone as a digital bunker

A hardware wallet secures your Bitcoin, but the smartphone you use to manage it can be a weak point. GrapheneOS turns your Android into a hardened operating system without Google telemetry, with app sandboxing and genuine data protection. The more secure the device in your hand, the more secure your entire Bitcoin infrastructure.

Volume 1: GrapheneOS: Android in the Age of Surveillance

alien-investor.org/en/buecher

ALIEN PHONE SERVICE

GrapheneOS on a Pixel, from €179, fully set up from €249, each plus device price

alien-investor.org/en/grapheneos-flash-service

On-Chain Privacy: If you have withdrawn Bitcoin from a KYC exchange, your coins are linked to your identity. CoinJoin can mathematically break this link, but it is an advanced topic with its own pitfalls. You will find an honest assessment with a clear warning in the bonus section "How free you really are" at the end of this guide.

UTXO management: Your wallet balance is not a single number but a set of individual "coins" (UTXOs), the amounts you received in each transaction. Handling them deliberately pays off. Very small UTXOs ("dust") can cost more to spend than they are worth when network fees are high, because spending alone eats up the fee. Very large, merged UTXOs, on the other hand, reveal more about your total holdings when spent and weaken your privacy. The sweet spot is medium-sized UTXOs combined with avoiding address reuse. Wallets like Sparrow show your UTXOs individually and let you select them deliberately (coin control).

How big should a UTXO be? As a rough rule of thumb a single UTXO should be worth at least around **1,000,000 sats (0.01 BTC)**. At that size the cost to spend it stays below roughly one percent of its value even at expensive network fees of about 100 sat/vB. The purely technical dust limit is far lower (around 300 sats for SegWit addresses), but amounts that small become practically unspendable when fees are high, because spending alone eats up the fee. Quiet markets with low fees are the right moment to merge many small UTXOs into a larger one (consolidation).

Multisig: For larger amounts it can make sense to combine multiple hardware wallets so that transactions require multiple signatures. A single compromised seed phrase is then not enough. Sparrow Wallet handles multisig well.

Books that go deeper

If you want to truly understand the foundation, there is no way around these books. Order directly via the QR code.

Golden Age — Florian Bruce-Boye ⭐ Top pick

amzn.to/49MtATq

One of the best books on Bitcoin and the monetary future. My clear reading tip when you want to grasp the big picture.

The Bitcoin Standard — Saifedean Ammous

amzn.to/4azT86q

The classic. Why hard money matters and what fundamentally sets Bitcoin apart from any fiat currency.

The Fiat Standard — Saifedean Ammous

amzn.to/3RZ396U

The sequel: how the fiat money system actually works and what price we all pay for it.

GOLDEN AGE
amzn.to/49MtATq

BITCOIN STANDARD
amzn.to/4azT86q

FIAT STANDARD
amzn.to/3RZ396U

The three links above are Amazon affiliate links. If you order through them, you support this channel at no extra cost to you.

Further resources: Bitcoin.org, Bitcoin Magazine, Andreas Antonopoulos' "Mastering Bitcoin" (more technical), as well as the documentation for BitBox02, Sparrow Wallet, and your node system.

Chapter 8: Recognizing and avoiding scams

Self-custody protects you from exchange hacks and bankruptcies. But there is one attack vector no hardware can defend: you yourself. Most Bitcoin is not lost through broken encryption, but because someone was deceived. Anyone who knows the following schemes will not fall for them.

> **NO ONE EVER ASKS FOR YOUR SEED PHRASE**
>
> No real support team, no wallet app, no employee, no tax office will ever ask for your 24 words or your passphrase. Whoever asks is a scammer. No exceptions. Never enter them anywhere except directly on your hardware wallet during a recovery.

The most common schemes

Fake support. You look for help in a forum, on X, or Telegram, and suddenly "support" messages you by direct message. Real providers never message first by DM. They also never send you to a page that asks for your seed phrase "for verification".

Phishing and cloned apps. Scammers build websites and apps that look deceptively like the real BitBoxApp or wallet site. Only download software from the official domain (bitbox.swiss) and verify the signature. Do not click wallet links from emails or ads, type the address yourself.

"Send 1 BTC, get 2 back". Fake giveaways, often with celebrity images or fake livestreams. There is no gift. Money you send is gone. Bitcoin transactions are irreversible.

Fake hardware wallets. Devices from third parties (Amazon, eBay, marketplaces) can be tampered with using a pre-installed seed phrase. Buy exclusively from the manufacturer and check the seal. A genuine wallet always generates the seed phrase itself, during initial setup, before your eyes.

> **CLIPBOARD MALWARE & ADDRESS POISONING**
>
> Malware can secretly swap a copied Bitcoin address in your clipboard for the attacker's address. With address poisoning, someone sends you tiny amounts from an address resembling your earlier ones, hoping you will copy it by mistake later. **Protection:** on every transaction, compare the first and last group of four characters

> of the address on your hardware wallet's display with what is on the screen. That is exactly what your device's own display is for, malware cannot manipulate it.

Fake "recovery services". Anyone who has lost coins quickly finds services promising "recovery" for an upfront fee. This is almost always a second scam. Lost seed phrases cannot be recovered, no one can do it.

> **RULES OF THUMB AGAINST FRAUD**
>
> Never enter or share your seed phrase digitally. Only get software from the official source. No one messages first by DM. If it sounds too good to be true, it is a scam. Under pressure or urgency: stop, that is a scammer's favorite weapon. When in doubt, trust no one and verify yourself.

Your final checklist

Go through this list before you lean back and relax. Each point protects you against real risks.

To print: This list is also available as a workshop sheet on a single A4 page, with boxes to tick and room for the date. Free on the guide page at alien-investor.org/en/bitcoin-einsteiger-guide.html.

01	I have bought Bitcoin on a trustworthy exchange or P2P platform.
02	I have ordered a BitBox02 directly from the manufacturer and received it sealed: original vacuum packaging with BitBox logo intact. An opened or damaged seal is a warning sign.
03	I have set up the wallet, written down the 24-word seed phrase by hand and practiced the recovery once.
04	I have secured the seed phrase on metal (or have a concrete plan to do so).
05	I store the seed phrase and hardware wallet in separate locations.
06	I have at least one second copy of the seed phrase at a different location.
07	I have sent a test amount to my hardware wallet and confirmed its arrival.
08	I have transferred the rest of my coins from the exchange to my hardware wallet.
09	I have not stored the seed phrase, any photo of it, or any screenshot anywhere digitally.
10	A trusted person knows that a seed phrase exists and what to do with it (or I have left written emergency instructions).

Anyone who can check off all ten points has laid the foundation for genuine Bitcoin sovereignty. What comes after is refinement.

Stay sovereign,

Alien Investor

Bonus: How free you really are

With self-custody you have achieved the most important thing: your Bitcoin truly belongs to you. Everything that follows is optional. But it is worth knowing what is possible, because that is exactly what shows how much freedom is in this money. I have explained the following topics in detail on the blog. Scan the QR code or type in the address.

UNDERSTAND

What Bitcoin really is — and why custody is everything

Limited supply, decentralization, real ownership: what sets Bitcoin apart from money in a bank account, in two compact articles. The second uses on-chain data to show why Bitcoin on an exchange loses its actual purpose.

PROPERTIES
alien-investor.org/en/bitcoin.html

SELF-CUSTODY
alien-investor.org/en/self-custody.html

PUT IN CONTEXT

The Bitcoin Power Law — a model, not a guarantee

The Power Law is a widely discussed observational model that describes Bitcoin's price history remarkably well along a trend line. Important: it is an empirical observation, not a price promise and not a guarantee for the future. The **Mission Control** dashboard shows the key financial data and the current price live, and the **Power Law chart** shows the trend line in real time.

POWER LAW EXPLAINED
alien-investor.org/en/bitcoin-power-law.html

LIVE CHART
api.alien-investor.org/charts.html?lang=en#powerlaw

MISSION CONTROL
api.alien-investor.org/mission-control-v2.html?lang=en

ASSESSING

How I assess the Bitcoin price

Is Bitcoin expensive or cheap right now? I do not approach this on gut feeling but with transparent tools: market cycles, on-chain data (such as MVRV and realized price), capital flows, sentiment and the macro environment. That does not produce a price target but a range: where would I add, where would I be cautious? In the article I show this step by step, so you can put price moves into context yourself instead of chasing the next hype. I publish regular Bitcoin price analyses on the blog at **alien-investor.org**.

ASSESS THE PRICE
alien-investor.org/en/wie-ich-den-bitcoin-kurs-bewerte.html

MARVEL

Solo mining vs. lottery — how decentralized Bitcoin is

You do not have to mine, and solo mining is not an income source: the total computing power of the network is enormous, so finding your own block with small hardware is extremely unlikely. But the fact that anyone can theoretically join in, without permission, without KYC, shows how decentralized Bitcoin really is. The article does the honest math against a lottery ticket.

MINING VS. LOTTERY

alien-investor.org/en/solo-bitcoin-mining-vs-lotto.html

LATER ONLY · WITH CAUTION

CoinJoin — privacy for your coins

CoinJoin breaks the link between the origin and destination of your Bitcoin. That is powerful, but not for beginners. **Only do this once you truly understand everything.** Mixing wrong is more likely to worsen your privacy than improve it.

The most important rule: mixed coins must never go back into the KYC world. Regulated exchanges recognize CoinJoin outputs and may reject such deposits, freeze your account, or demand a full proof of origin. So never send mixed coins to a KYC exchange, treat them as a separate, KYC-free pot.

Legally, in Germany: the private use of your own, legally acquired coins is not illegal, only operating a mixing service is the sensitive activity. The tool landscape is in flux (Samourai shut down, Wasabi coordinator discontinued). For getting started, self-custody and buying noKYC provide by far the biggest privacy lever, at far less risk.

No liability: none of this is legal or tax advice. You act on your own responsibility and bear full responsibility for your assets. Read the articles before you mix anything.

UNDERSTAND COINJOIN

alien-investor.org/en/coinjoin-fungibility.html

PRACTICE: ON TAILS

alien-investor.org/en/coinjoin-tails-praxis.html

WASABI RESEARCH

alien-investor.org/en/wasabi-wallet-research-transparent.html

MAXIMUM PRIVACY

Swapping Bitcoin to Monero — pay privately without a trace

One thing up front: we are Bitcoiners, and Bitcoin remains our first choice, as hard money and as a long-term store of value. Still, we do not want to leave this option unmentioned. Bitcoin is transparent: every transaction sits in the blockchain forever, out in the open. For private everyday payments you can swap noKYC Bitcoin peer-to-peer (for example as an atomic swap) into Monero, a currency where amount, sender and recipient are hidden by default. That way you pay without exposing your entire Bitcoin history. You will not find this route in ordinary beginner guides. Alien Investor deliberately goes deep here and tells the whole story, no half measures, without ever touching fiat.



MONERO WITHOUT FIAT

alien-investor.org/en/monero-ohne-fiat.html

Glossary

The key terms from this guide, briefly explained.

Sat (Satoshi)

Smallest unit of Bitcoin. One Bitcoin = 100,000,000 sats. Named after its creator Satoshi Nakamoto.

Seed phrase

Sequence of usually 24 words from which your entire wallet is derived. Whoever has it has the coins. Never store it digitally.

Passphrase (25th word)

Optional, self-chosen string in addition to the seed phrase. Creates a separate, hidden wallet.

Self-custody

Holding your own keys instead of entrusting them to an exchange.

Hardware wallet

Small device that generates and stores your private key offline. The key never leaves the device.

Private key / public key

The private key proves ownership and signs transactions (secret). The public key, or address, may be shared to receive Bitcoin.

KYC / noKYC

KYC ("Know Your Customer") = buying with identity verification at regulated exchanges. noKYC = buying person to person without identity verification.

UTXO

"Unspent Transaction Output", an unspent amount of Bitcoin at an address. Your balance is the sum of your UTXOs, like individual coins and notes in a wallet.

Mempool

Waiting area for transactions not yet confirmed in a block. Viewable on mempool.space.

Confirmation

A transaction is confirmed once it lands in a block (about every 10 minutes). Each additional block is another confirmation.

sat/vByte

Unit of the network fee: satoshi per virtual byte. Determines how quickly a transaction confirms, regardless of the amount sent.

Native SegWit (bc1q) / Taproot (bc1p)

Modern Bitcoin address formats. bc1q is the recommended standard with low fees, bc1p (Taproot) is the newest standard.

Full node

Your own copy of the Bitcoin blockchain that verifies all rules itself. Maximum sovereignty, you trust no one.

Lightning Network

Second layer on top of Bitcoin for fast, cheap micro-payments, such as zaps on Nostr.

Cold storage / air-gap

Storage without an internet connection. An air-gap medium (e.g. the BitBox MicroSD) never touches a networked computer.

Resources & Contact

Direct Links

BitBox02 — Hardware Wallet

Order directly from the manufacturer. Bitcoin-only Edition recommended. Receive sealed, check vacuum packaging with BitBox logo.

5 % discount with code **ALIENINVESTOR**

alien-investor.org/bitbox

21bitcoin — Buy Bitcoin in Europe

Bitcoin-only app from Europe. Instant purchases and DCA savings plans, ideal for stacking sats regularly.

Permanent 0.2 percentage point fee reduction with code **ALIENINVESTOR**

alien-investor.org/21bitcoin

BitBox02 Nova — for Apple users (iPhone/iPad)

Connects via Bluetooth to the BitBoxApp, fully iOS-compatible. Same security architecture as the BitBox02, open-source, made in Switzerland.

5 % discount with code **ALIENINVESTOR**

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Where to go from here

Everything below is my own offering. I mention it because it picks up exactly where this guide stops.

The book series

Volume 1, GrapheneOS: Android in the Age of Surveillance. The smartphone as a secure foundation.
Volume 2, Digital Sovereignty: The Complete System. Privacy architecture for computer, network and everyday life.
Volume 3, The Decentralized System: Bitcoin. Stocks. True Ownership. Bitcoin, stocks and precious metals in true ownership, taxes included.
Volume 4, Your Own Bank: Master Bitcoin Self-Custody. Your own seed, multisig, inheritance: custody for advanced users.
As a Kindle edition on Amazon or DRM-free through direct sale. Direct buyers get every new edition for free.

alien-investor.org/en/buecher.html

Our apps — free and offline

Alien Pass, Sachwert-Tresor, Ausgaben-Tracker, Alien Fitness, Alien Notes and the BTC Steuertool, built for Android and GrapheneOS. All of them run offline, are open source, and are free, with no ads, no subscription, and no price tag. Download sources and signing fingerprints are on the overview page.

alien-investor.org/en/apps.html

Personal help

Setup Review (€49): You describe your Bitcoin setup and get a written assessment with weak points and improvements.
Alien Phone Service: GrapheneOS on a Pixel: from €179 as a clean system, fully set up from €249, each plus device price.

alien-investor.org/en/setup-review.html · alien-investor.org/en/grapheneos-flash-service

Did this guide help you?

It stays free. If you want to give something back: share it, follow the channel, or donate via Lightning, Bitcoin onchain, bank transfer or card.

alien-investor.org/en/spenden

Questions

You have a specific question not answered in the text, or you are stuck at a certain point? Feel free to write. I will answer as soon as I can, but with a free guide I cannot promise a fixed response time. If you want a thorough review of your whole setup, the Setup Review is the right place.

CONTACT

kontakt@alien-investor.org
Subject: Bitcoin Guide Question
Briefly describe where you are and what is unclear.

Disclaimer

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